

1. What is the `sizeof` Operator?

The **`sizeof` operator** is used to find the **size of a data type or variable in bytes**.

Syntax

```
sizeof(variable)
```

or

```
sizeof(data_type)
```

The result of `sizeof` is given in **bytes**.

2. Example: `sizeof` with Data Types

```
#include <iostream>
using namespace std;

int main() {
    cout << sizeof(int) << endl;
    cout << sizeof(float) << endl;
    cout << sizeof(double) << endl;
    cout << sizeof(char) << endl;

    return 0;
}
```

Possible Output:

```
4
4
8
1
```

Note: The size of some data types can vary depending on the compiler and computer system. `char` is always 1 byte.

3. Example: `sizeof` with Variables

```
#include <iostream>
using namespace std;
```

```
int main() {  
    int age = 20;  
    double salary = 50000.5;  
    char grade = 'A';  
  
    cout << sizeof(age) << endl;  
    cout << sizeof(salary) << endl;  
    cout << sizeof(grade) << endl;  
  
    return 0;  
}
```

Possible Output:

```
4  
8  
1
```

Understand the Code

- `sizeof(age)` → finds the size of the `int` variable.
- `sizeof(salary)` → finds the size of the `double` variable.
- `sizeof(grade)` → finds the size of the `char` variable.

4. `sizeof` Does Not Give the Value

Consider:

```
int age = 20;  
  
cout << age << endl;  
cout << sizeof(age) << endl;
```

Output:

```
20  
4
```

Here:

- `age` gives the **value** → 20
- `sizeof(age)` gives the **memory size** → 4 bytes on a typical system.

5. `sizeof` with Arrays

`sizeof` can also be used with arrays.

```
#include <iostream>
using namespace std;

int main() {
    int numbers[5];

    cout << sizeof(numbers);

    return 0;
}
```

If an `int` is 4 bytes:

$5 \times 4 = 20$ bytes

Output:

20

6. Finding the Number of Elements in an Array

We can use `sizeof` to find the number of elements in an array:

```
int numbers[5];

int total = sizeof(numbers) / sizeof(numbers[0]);

cout << total;
```

Output:

5

Why?

```
sizeof(numbers)    = 20 bytes
sizeof(numbers[0]) = 4 bytes

20 / 4 = 5
```

Therefore, the array contains **5 elements**.



Quick Revision

Expression	Purpose
<code>sizeof(int)</code>	Size of <code>int</code>
<code>sizeof(float)</code>	Size of <code>float</code>
<code>sizeof(x)</code>	Size of variable <code>x</code>
<code>sizeof(array)</code>	Total size of array
<code>sizeof(array) / sizeof(array[0])</code>	Number of array elements

Remember

`sizeof` tells us how many bytes are occupied by a data type, variable, or object.

Important: `sizeof` returns a size in **bytes**, not bits.